

INSTITUTE OF ACTUARIES OF INDIA

EXAMINATIONS

20th December 2018

Subject CT3 – Probability & Mathematical Statistics

Time allowed: Three Hours (10.30 – 13.30 Hours)

Total Marks: 100

INSTRUCTIONS TO THE CANDIDATES

1. *Please read the instructions inside the cover page of answer booklet and instructions to examinees sent along with hall ticket carefully and follow without exception.*
2. *Mark allocations are shown in brackets.*
3. *Attempt all questions, beginning your answer to each question on a separate sheet. However, answers to objective type questions could be written on the same sheet.*
4. *Please check if you have received complete Question Paper and no page is missing. If so, kindly get new set of Question Paper from the Invigilator.*

AT THE END OF THE EXAMINATION

Please return your answer book and this question paper to the supervisor separately. You are not allowed to carry the question paper in any form with you.

- Q. 1)** The table below summarizes the marks (X) scored, out of a maximum of 100, by students in a mathematics examination.

Marks	Boys	Girls
$X < 40$	13	27
$40 \leq X < 60$	48	52
$60 \leq X < 80$	27	15
$80 \leq X$	14	4

- i) Plot the cumulative distribution curve for the marks scored by students (both girls and boys combined) assuming that the marks are distributed uniformly within each interval. (2)
- ii) Hence, calculate the mean and median marks. (3)
- [5]

- Q. 2)** Four dice are thrown simultaneously. Calculate the probability that sum of numbers is exactly 19. [4]

- Q. 3)** The probability mass function of a random variable X is given by:

$$P(X = x) = \begin{cases} C(2x - 1) & ; x = 1, 2, \dots, 10 \\ 0 & ; \text{otherwise} \end{cases}$$

- i) Find the value of C . (1)
- ii) Calculate the expected value of X . (3)
- [4]

- Q. 4)** On New year's eve 2018, which falls on Sunday, a person resolves to provide charity in the next month as under:

On 1st January 2018, he will provide free lunch to one orphan; on 2nd January to two other orphans; on 3rd January to three other orphans and so on until 31st January 2018. In addition to this charity he will also donate a pair of clothes to each of them, if it is a Sunday.

- i) Calculate the probability that a randomly selected orphan receiving charity will get lunch and clothes. (3)
- ii) Calculate the probability that a randomly selected orphan will be served on third Sunday, given that the person got lunch and clothes. (2)
- [5]

- Q. 5)** The moment generating function (mgf) of a random variable X is given by:

$$M_x(t) = \sum_{k=0}^{\infty} \frac{e^{kt-1}}{k!}.$$

Find $P(X = 3)$. [2]

- Q. 6)** Let X_1 and X_2 be two i.i.d. random variables having the probability mass function

$$P(X_1 = i) = \begin{cases} \frac{1}{4} & ; i = 1, 2, 3, 4. \\ 0 & ; \text{otherwise.} \end{cases}$$

Using their *mgf*s or otherwise, find the probability mass function of $X_1 + X_2$. [4]

- Q. 7)** Let Y and Z be discrete (integer-valued) random variables with joint probability mass function

$$p(y, z) = \begin{cases} \frac{1}{36} & \text{for } 1 \leq y = z \leq 6 \\ \frac{2}{36} & \text{for } 1 \leq y < z \leq 6 \\ 0 & \text{otherwise.} \end{cases}$$

Verify whether Y and Z are independent. [5]

- Q. 8)** The probability density function of X is:

$$f(x) = \frac{e^{-x}}{(1 + e^{-x})^2}; -\infty < x < \infty$$

Find the probability density function of $\frac{e^X}{1+e^X}$ [5]

- Q. 9)** Light bulbs are installed into sockets. Assume that each bulb has mean life of 3 months with a standard deviation 0.25 month and their lifetimes are independent. How many bulbs should be bought so that we can be 95% sure that supply of these bulbs will last for at least 50 months? [6]

- Q. 10)** The time taken (X) by an aeronautical company to build an airplane is a normal random variable with mean 1000 man-hours and standard deviation of 240 man-hours. The company promises to build one airplane by taking at most 1100 man- hours and agrees to pay a penalty of Rs 1, 00,000 for each man- hour spent in excess of 1100.

i) Calculate the probability that company pays a penalty of at least Rs 20, 00,000. (3)

ii) Let $X_1, X_2, \dots, X_n$ be a random sample of times taken to build 'n' airplanes. The 73.5th percentile of the statistics $\sum_{i=1}^n \frac{(X_i - 1000)^2}{240^2}$ is 10. Determine the sample size n . (4)

[7]

- Q. 11) i)** Show that the method of moments estimate for 'a' of a continuous uniform distribution $U(a, b)$ is $\hat{a} = \bar{x} - \sqrt{3}s$ where $\bar{x}$ is sample mean and s is sample standard deviation. (3)

ii) State a formula for $\hat{b}$ (method of moments estimate for 'b') in terms of sample mean and sample standard deviation. (1)

iii) Create a sample of five observations and use the sample to demonstrate potential weakness of the method of moment's estimation of 'a' and 'b' for $U(a, b)$. (4)

iv) Find the maximum likelihood estimate of b for $U(a, b)$ based on a sample $x_1, x_2, \dots, x_n$, when $a = 0$. (3)

[11]

Q. 12) ABC Space agency, responsible to protect a planet from asteroid collision, developed new space-to-space missiles to be loaded in satellites.

The agency planned missile trial in two steps for testing $H_0: p = 0.1$ versus $H_1: p > 0.1$, where p is the proportion of hits of missiles, each missile targeted at similar asteroid.

At the first step, 12 missiles will be fired. If three or more independent hits are observed among the (first) 12 missiles, H_0 is rejected, the study is terminated, and no more missiles are fired.

Otherwise, another 12 missiles will be fired in the second step. If a total of five or more independent hits are observed among the 24 missiles fired in the two steps, then H_0 is rejected.

i) Calculate the probability of Type I error for the two step testing procedure. (4)

ii) Calculate the probability of rejecting the null hypothesis H_0 when $p = 0.3$. (3)

iii) Calculate the probability of Type II error when $p = 0.3$. (1)

Suppose that 10 Space agencies have developed missiles similar to ABC Space agency. They have performed only the first step trials and observed an aggregate of 40 hits out of 120 independent missiles fired.

iv) Calculate an approximate lower bound of 90% right-tailed confidence interval for 'p'. (show up to four decimal places) (3)

v) Test the hypothesis that $H_0: p = 0.278$ against $H_1: p > 0.278$ at 10% level of significance. (3)

vi) Comment on your results of confidence interval obtained in (iv) and hypothesis testing in (v). (3)

Aptitude tests were conducted by ABC Space agency at their two Institutes (1 and 2) that provide special training on missile technology. The test scores are shown in the table below:

	Institute 1	Institute 2
No. of trainees	12	15
Mean scores	65	70
Standard deviation	54	70

vii) Test the equality in mean scores of the populations associated with the two Institutes. State any assumptions made. (4)

[21]

- Q. 13) i)** Determine the MLE of α and β under full normal model $Y_i \sim N(\alpha + \beta x_i, \sigma^2)$.
(You are not required to verify that your answer corresponds to a minimum /maximum) (5)

- ii)** It is well known that large bodies of water have a lessening effect on the temperature of the surrounding land-masses. On a cold night, temperatures (in centigrade) were recorded, as shown in the table below, at equal distances along a cross section running downwind from a large lake.

Site (x)	1	2	3	4	5	6	7	8	9	10
Temp.(y)	27.0	25.0	23.4	22.6	21.8	21.4	21.2	21.0	20.8	21.0

$$\bar{x} = 5.5; S_{xx} = 82.5$$

Notice that the temperatures drop rapidly and then level off as we move away from the lake.

Fit an appropriate linear regression model for this data, by making a suitable transformation. (6)

- iii)** There is a petrol bunk in the middle of sites 4 and 5. Calculate 95% confidence interval for the predicted temperature at that petrol bunk ($x = 4.5$). (5)
[16]

- Q. 14)** Estimate the parameters μ and τ_i using least squares method for the model:

$$Y_{ij} = \mu + \tau_i + e_{ij}, i = 1, 2, 3, \dots, k; j = 1, 2, 3, \dots, n_i,$$

where $e_{ij} \sim N(0, \sigma^2)$ are independent random variables;

μ is overall population mean;

τ_i is the deviation of the i^{th} treatment mean from μ and $\sum_{i=1}^k n_i \tau_i = 0$.

(You are not required to verify that your answer corresponds to a minimum /maximum.) [5]
